

GHAZAL JOHRI

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SUMMARY

Computer Science undergraduate specializing in applied AI and backend development, with experience building RAG systems, agent-based workflows, and LLM-powered applications. Designed Elena — a semi-autonomous AI agent with Human-in-the-Loop (HITL) controls, real-time RFP discovery via web crawling, live meeting attendance, and virtual desktop navigation — at Centific's 2-week residential hackathon. Backend engineering intern with measurable API performance wins. Targeting AI/ML and software engineering roles where systems thinking and fast execution matter.

EDUCATION

B.Sc. Data Science & Applications - IIT Madras 2023 – Present (Concurrent)

B.Tech Computer Science & Engineering - Amity University, Noida 2022 – 2026 (Concurrent)

Dual-degree: pursuing both programmes simultaneously. Core coursework: DSA, OS, DBMS, Machine Learning, NLP.

WORK EXPERIENCE

AI Engineering Intern · Centific Oct 2025 – Apr 2026

- Engineered HITL validation workflows for enterprise LLM pipelines, improving model output compliance by flagging edge-case failures before reaching production.
- Built model evaluation infrastructure — custom metrics, scoring dashboards, and feedback loops — used directly in production QA cycles across AI feature teams.
- Collaborated with senior ML engineers on multi-modal data ingestion and model deployment pipelines, gaining hands-on familiarity with

Backend Engineering Intern · Sekyoor Solutions Pvt. Ltd. May 2025 – Jul 2025

- Reduced API response latency by **20%** by rewriting N+1 SQL queries and introducing a Redis-style in-memory cache layer.
- Designed and shipped 6+ RESTful endpoints with JWT authentication, RBAC, and input validation from scratch; zero security incidents in production.

PROJECTS

Elena - Autonomous RFP Discovery Agent | Python · LLMs · Web Crawling · RAG · HITL · Desktop Automation · Meeting APIs

- System architecture:** built a fully-pipelined agentic system — web crawler → HTML parser → LLM relevance classifier → confidence scorer → HITL review queue → stakeholder communication — operating end-to-end without manual intervention.
- RFP discovery engine:** designed a web-crawling layer that autonomously discovers and ingests RFP documents from public procurement portals, then ranks opportunities by a confidence score derived from LLM semantic matching against company capability profiles.
- HITL control layer:** implemented a Human-in-the-Loop approval workflow where procurement managers review, approve, or reject shortlisted RFPs before Elena executes follow-up actions — preserving human oversight in a semi-autonomous loop.
- Humanoid presence:** gave Elena a dedicated team email identity, the ability to join and participate in live video meetings, and a conversational AI persona — enabling her to communicate findings to stakeholders without human mediation.
- Virtual desktop navigation:** engineered OS-level automation enabling Elena to launch browsers, navigate procurement websites, fill forms, and extract RFPs on a live virtual desktop — extending capability far beyond API-only bots.
- Live screen-sharing:** integrated real-time screen broadcast so Elena can present her discovery pipeline and confidence-scored results live inside meetings, enabling transparent AI decision-making.

Agentic AI Task Orchestrator | Python · LangChain · LLMs · RAG · Tool Use · Async APIs

- Designed a multi-agent orchestration framework where a planner LLM decomposes goals into subtasks, dispatches them to specialist sub-agents, and synthesises final output — enabling autonomous multi-step reasoning without human prompting.
- Integrated a RAG memory layer backed by a vector store: agents embed and retrieve domain knowledge at runtime, grounding every action in retrieved context and reducing hallucination rate in evaluated test cases by ~35%.

Production-Grade RAG Pipeline | Python · FAISS · OpenAI / HuggingFace Embeddings · FastAPI

- ▶ Engineered a full RAG pipeline: PDF/text ingestion → semantic chunking → embedding generation → FAISS vector indexing → hybrid dense+sparse retrieval → re-ranking → grounded LLM generation with inline source attribution.
- ▶ Deployed as a FastAPI microservice; achieved sub-300 ms median retrieval latency on a 10k-document corpus in local benchmarking.
- ▶ Cut hallucination rate in domain-specific QA tasks by anchoring every generated response to retrieved chunks, validated through manual evaluation on a 100-question test set.

Interview AI Guide | MERN Stack · Gemini API · NLP · Adaptive Questioning Engine

- ▶ Architected an AI-powered mock interview platform: users select domain and seniority; Gemini dynamically generates, evaluates, and follow-up questions in real time based on the candidate's last answer.
- ▶ Built adaptive difficulty logic — questions escalate or de-escalate based on a running confidence score computed from response quality signals — simulating a real technical screen.
- ▶ Authoring academic research paper on the system's NLP methodology and evaluation results; targeting publication in a peer-reviewed technical journal.

TECHNICAL SKILLS

AI / ML: LLMs, Agentic AI Systems, RAG, HITL Workflows, Transformers, NLP, Embeddings, ML Classification, Tool Use

Languages: Python, JavaScript (ES6+), Java, C++

Frameworks: LangChain, FastAPI, Express.js, React, Node.js

Databases: PostgreSQL, MySQL, FAISS, Pinecone (vector), Redis (caching)

Tools: Git, Hugging Face Hub, Web Crawling, Desktop Automation (Pyautogui / Playwright), REST API design

Concepts: Data Structures & Algorithms, OOP, System Design, Modular Architecture, OWASP Security Practices

ACHIEVEMENTS

- ▶ **Centific Premier Hackathon 2.0** — Selected for 2-week residential AI hackathon out of competitive applicant pool; built Elena, a fully-pipelined autonomous RFP agent with HITL controls, virtual desktop nav, and live meeting participation.
- ▶ **SheCodes Foundation Scholarship for Women in Tech** — Merit-based award recognising advanced aptitude in software engineering.
- ▶ **Research Paper (In Progress)** — First-authored paper on Interview AI Guide's adaptive NLP evaluation methodology; targeting peer-reviewed publication.